Literature Highlights Report Chua Jia Kang Eddy (02444183)

Importance of SSI

1.AlterEgo: A Personalized Wearable Silent Speech Interface [1]

Paper Summary

This paper introduces AlterEgo, a wearable silent speech human-machine interface. Following a rise in popularity of natural user interfaces and speech input, speech technologies like virtual assistants (e.g. Siri) have been integrated in our daily lives. However, such voiced speech recognition systems present limitations such as privacy concerns and background interference. AlterEgo addresses these challenges by leveraging internal speech, which is the act of speaking to oneself mentally, enabling seamless, private interaction with computing devices without audible sound or visible muscle movement.

AlterEgo consists of a wearable mask design wrapped around the back of the head, with electrode extensions placed on 7 target facial regions. Neuromuscular signals captured by these electrodes undergo multiple stages of processing, including filtering and transformations, before being classified by a silent speech recognition model, which is a 1-dimensional convolutional neural network, into word labels. The recognized phrases are interpreted based on the user's intended application and the system provides auditory output via bone conduction headphones. An example application would be issuing reminders and scheduling tasks at specific times. The model was evaluated through a multi-user study with 10 participants, and the model achieved a promising average classification accuracy of 92.01% between all users.

Key Relevance obtained from this literature

The importance of a silent speech interface is emphasized, while the limitations of voiced speech recognition such as privacy and noise are acknowledged

Infrared sensing

2. Muscle activity measurement using visible light and infrared [2]

Paper summary

This paper proposes MAX30105, an optical sensor module with infrared and visible LEDs and a photodiode, as a potential alternative methodology to EMG for the measurement and detection of muscle activity. A series of tests were conducted using the sensor to verify the efficacy of this approach.

With the aid of an elastic band, the optical sensor and an EMG sensor were firmly positioned adjacently on the forearm of a participant to detect movements of the ring finger. A strong relationship was found between the EMG signal and the recorded signal from infrared and red-light emissions. The relationship between muscle movement and the recorded optical signal was further validated by comparing it with angular velocity data from a gyroscope placed on the moving finger. To demonstrate the comprehensiveness of the sensor, additional experiments for other human locomotion were conducted. The photodiode current changes during wrist and forearm movements were both aligned to the expected results, and a test on finger movement on a larger sample size verifies the consistency of the system across users. Overall, the findings demonstrate a strong link between the EMG signal and finger, wrist, and forearm movement, confirming the viability of employing the MAX30105 as an EMG substitute.

Key Relevance obtained from this literature

Our project requires selecting a sensing modality to detect laryngeal muscle movement during silent speech. This paper demonstrates that near-infrared light is effective for monitoring muscle activity. It also highlights key advantages of infrared over other wavelengths, including higher sensitivity and deeper tissue transparency.

Data processing and analysis pipeline

3. SpeeChin: A smart Necklace for silent speech recognition [3]

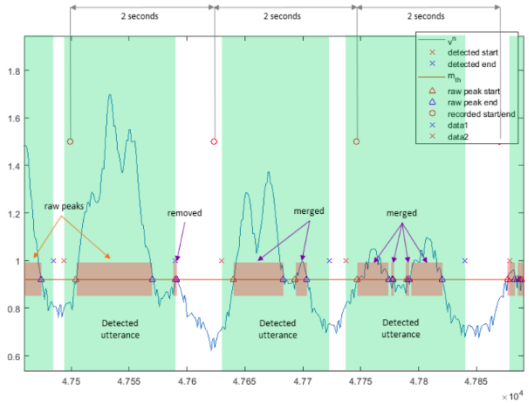
Paper Summary

This paper introduces SpeeChin, a smart necklace capable of detecting 54 English and 44 Chinese silent speech commands. It uses an infrared camera housed in the necklace to capture variations in the shape of the neck and face for silent speech interpretation.

The system follows a multi-stage pipeline. An OV5647 image sensor records reflected light, and image pre-processing techniques are applied to the data, including: (1) Otsu binarization and H channel histogram to identify the region of interest. (2) affine transformation to rectify shifts in device angle and positioning during head motion. (3) differential image generation to highlight subtle facial movement changes. For utterance detection, the following steps are applied: (1) the system calculates the vigorosity from differential images- defined as the standard deviation of all pixel intensities.

$$v_i = \mathrm{std}\left(\frac{1}{3}\sum_{k=-1}^{1} I_{i+3+k} + \frac{1}{3}\sum_{k=-1}^{1} I_{i-3+k} - \frac{2}{3}\sum_{k=-1}^{1} I_{i+k}\right)$$

(2) A 4-second moving average smoothes these values and these values are plotted over time (3) Peaks (> threshold V_{th}) are identified, lower vigorosity values are discarded as non-speech (4) Peaks too close in time (<t_m) are merged.



The pre-processed data is input into Speech Net, an end-to-end CRNN model, which combines CNN (for spatial features) and RNN (for temporal features) to classify silent commands.

Evaluation across two user studies reported high accuracies: 90.5% (English) and 91.6% (Chinese).

Key relevance obtained from this literature:

1) The feasibility of using an infrared camera with 850nm light for detecting silent speech has been demonstrated
2) The Pre-processing techniques and pipeline can be adopted or adapted
3) A CRNN model has been shown to be effective in handling infrared camera data

Database implementation

4) Montreal Forced Aligner: trainable text-speech alignment using Kaldi[4]

Paper Summary

This paper introduces an open-source command line tool that can automatically align words and phonemes given audio recording and its corresponding transcript. Its architecture has led to several important characteristics: (1) Captures contextual diversity in phone realization using triphone acoustic models. (2) Adapts to interspeaker variabilities by the feature space Maximum Likelihood Linear Regression method (3) Trainability (4) Robustness to unrecognized words

The performance of the system was evaluated by computing the absolute difference between word and phoneme boundaries found by MFA and manual annotation across two representative corpora in linguistics. Results showed a mean alignment error of both word and phoneme boundaries to be on par with human annotators, affirming MFA's alignment quality. Compared to existing popular aligners PLA and FAVE, MFA had the lowest mean and median alignment errors across two corpora datasets and both boundary categories, proving that its complex architecture had a positive effect on the recognition accuracy. Furthermore, comparing mean and median alignment errors between an untrained and trained version on the dataset to be aligned, there was only one case the retrained model led to distinct better performance, acknowledging that MFA can work well out of the box.

Key relevance obtained from this literature

A reliable dataset is essential for data analysis. For our silent speech interface to map to any text, we require a phoneme library. This paper demonstrates that MFA is a promising tool to identify phoneme boundaries in speech. Once these boundaries are established, a simple script can be written to segment speech into individual phoneme-level samples.

References

[1] A. Kapur, S. Kapur, and P. Maes, “AlterEgo: A personalized wearable silent speech interface,” in *Proc. 23rd Int. Conf. on Intelligent User Interfaces (IUI)*, Mar. 2018, pp. 43–53.

[2] M. Sikora and S. Paszkiel, “Muscle activity measurement using visible light and infrared,” *IFAC-PapersOnLine*, vol. 52, no. 27, pp. 329–334, 2019.

[3] R. Zhang, M. Chen, B. Steeper, Y. Li, Z. Yan, Y. Chen, S. Tao, T. Chen, H. Lim, and C. Zhang, “SpeeChin: A smart necklace for silent speech recognition,” *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 5, no. 4, Art. no. 192, pp. 1–23, Dec. 2021, doi: 10.1145/3494987.

[4] M. McAuliffe, M. Socolof, S. Mihuc, M. Wagner, and M. Sonderegger, “Montreal forced aligner: Trainable text-speech alignment using Kaldi,” in *Proc. Interspeech*, 2017, pp. 498–502.